- *Elastic Stack*—While the Elastic Stack (Elasticsearch, Logstash, Kibana) can be used as an open source solution, managed services like Elastic Cloud provide additional support and features at a cost. This stack is excellent for searching, analyzing, and visualizing log data.

8.3 Troubleshooting LangChain.js

LangChain.js serves as an abstraction designed to simplify the development of applications utilizing LLMs in a composable manner. However, this added layer of abstraction can increase complexity and introduce new challenges when attempting to resolve unexpected errors.

NOTE The source code for the examples in this chapter is available at https://github.com/Generative-AI-Web-Apps/Code/tree/main/ch08/chat-langchain-troubleshooting.

Project code: chat-langchain-troubleshooting

The code for this example project is in the ch08/chat-langchain-troubleshooting folder. You can use that example by running this command in the root folder of the project repository:

```
$ npm run dev -w ch08/chat-langchain-troubleshooting
```

You'll also need either the OpenAI API or the Google API keys to access their REST API. The appendix offers instructions on how to set up a secret key for OpenAI and other providers:

```
// .env
OPENAI_API_KEY=<OPEN_AI_SECRET_KEY>
GEMINI_API_KEY=<GOOGLE_AI_SECRET_KEY>
```

NOTE The core LangChain.js library is implemented in TypeScript/JavaScript and does not require Python to be installed on your system. It runs natively in Node.js or browser environments. However, some community integrations or example projects may depend on Python tools or libraries and thus require a Python environment.

Figure 8.7 illustrates the two main troubleshooting paths when working with LangChain.js: chain execution problems and model integration problems.

TIP To troubleshoot problems with frameworks like Next.js, the Vercel AI SDK, and LangChain.js, combine technical knowledge, intuition, and hands-on experimentation. Problems often originate from multiple layers of the stack. Start by documenting your current state, reproducing the problem